

SECOND SEMESTER EXAMINATIONS 2025/26

Mock Question Paper Solution

TIME ALLOWED : One hour forty minutes

NAME OF CANDIDATE
STUDENT ID

THIS PAPER MUST NOT BE REMOVED FROM THE EXAMINATION ROOM

1. Find the derivative of $f(t) = \sin^3(t)$ with respect to t .

- ☐ A. $\cos(t)$
- ☐ B. $\sin(t)$
- ☒ C. $3\sin^2(t)\cos(t)$
- ☐ D. $3\sin(t)$

2. Given two vectors $\bar{X} = (11, 20, -1)^\top$ and $Y = (-3, 21, 10)^\top$, $\bar{Z} = \bar{X} + 2\bar{Y}$, What will be the value of $\bar{Z}$?

5 marks

- ☐ A. [5, 50, 12]
- ☒ B. [5, 62, 19]
- ☐ C. [3, 62, 12]
- ☐ D. [1, 29, 30]
- ☐ E. [4, 5, -1.5]

Solution $(11, 20, -1)^\top + (-6, 42, 20)^\top = (5, 62, 19)^\top$

3. Given the matrix $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ What are the eigenvalues of matrix A?

- ☒ A. 3 & 1
- ☐ B. 2 & 2
- ☐ C. 4 & 0
- ☐ D. 1 & -1

Solution $(2 - \lambda)(2 - \lambda) - 1 = 0, \lambda^2 - 4\lambda + 3 = 0, (\lambda - 3)(\lambda - 1) = 0$

4. A multiclass classifier is evaluated on 3 classes: A, B, and C. The F1-scores for each class are as follows:

F1-score for class A: 0.8

F1-score for class B: 0.6

F1-score for class C: 0.9

What is the macro-averaged F1-score for the classifier?

- ☐ A. 0.70
- ☒ B. 0.76
- ☐ C. 0.80
- ☐ D. 0.85

Solution The macro-averaged F1 score is $(0.8 + 0.6 + 0.9)/3 = 0.76$

5. Compute the L_∞ norm of the vector $(-3, 7, 2, 9)$

- ☐ A. 3
- ☐ B. 7
- ☒ C. 9
- ☐ D. 21

6. A dataset contains 1000 records. The attribute "Age" has missing values in 150 records. You decide to remove all records with missing values in "Age" before applying a data mining algorithm.

What percentage of the dataset will be removed?

- ☐ A. 10%
- ☒ B. 15%
- ☐ C. 20%
- ☐ D. 25%

7. You are given 4 points in 2D space:

A: (1, 1)

B: (2, 1)

C: (5, 4)

D: (6, 4)

Using Euclidean distance and Single Linkage Agglomerative Clustering, which two points will be merged first?

- ☒ A. A & B
- ☐ B. B & C
- ☐ C. C & A
- ☐ D. D & A

Solution $A - B = \sqrt{(21)^2 + (11)^2} = \sqrt{1} = 1$

$C - D = \sqrt{(65)^2 + (44)^2} = \sqrt{1} = 1$

$A - C \approx \sqrt{(51)^2 + (41)^2} = \sqrt{25} = 5$

$B - C \approx \sqrt{(52)^2 + (41)^2} = \sqrt{18} \approx 4.24$

The algorithm will choose either $A - B$ or $C - D$.

8. Given the matrix $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 0 & 1 \end{pmatrix}$ What is the rank of Matrix A?

☐ A. 1
☒ B. 2
☐ C. 3
☐ D. 4

Solution To find the rank, we look for the number of linearly independent rows (or columns).

Row 2 = 2 × Row 1 → Not independent

Row 3 is not a scalar multiple of Row 1 → Independent

So we have 2 linearly independent rows → Rank = 2

9. A data point P belongs to cluster A. The average distance from P to all other points in the same cluster A is $a = 2.0$

The average distance from P to points in the nearest neighboring cluster B is $b = 5.0$.

What is the Silhouette Coefficient for point P?

☐ A. 0.25
☐ B. 0.40
☒ C. 0.60
☐ D. 0.80

Solution $\frac{b-a}{\max\{a,b\}} = \frac{3}{5} = 0.6$

10. In a text classification task using the Naive Bayes algorithm, you are given the following:

The word "apple" appears 3 times in documents labeled Fruit.

The total number of words in Fruit documents is 100.

The vocabulary contains 20 unique words.

Using Laplace smoothing, what is the smoothed probability of the word "apple" given the Fruit class?

☒ A. 0.03
☐ B. 0.02
☐ C. 0.01
☐ D. 0.05

Solution $P(\text{apple}|\text{Fruit}) = \frac{3+1}{100+20} = 0.033$

11. A logistic regression classifier outputs the probability of class 1

Given:

Feature vector $x = [2, 1]$

Weights $w = [0.5, -1]$

Bias $b = 0.5$

What is the predicted probability $P(y = 1|x)$

☐ A. 0.27

☐ B. 0.38

☒ C. 0.62

☐ D. 0.73

Solution $z = 0.52 + (1)1 + 0.5 = 1.01 + 0.5 = 0.5$
 $P(y = 1|x) = \frac{1}{1+e^{-0.5}} \approx 0.622$